

Shrub encroachment enhances the infiltration capacity of alpine meadows by changing the community composition and soil conditions

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ABSTRACT

Due to global changes and anthropogenic disturbances shrubs are widely encroaching into alpine meadows, which may have profound impacts on the ecohydrological functioning of these key ecosystems. This phenomenon is particularly alarming in the alpine meadows of the Qinghai-Tibetan Plateau, the headwaters for many of Asia's largest rivers. However, little is known about the extent in which shrub encroachment alters the infiltration capacity of those soils. Here we examined the effects of shrub encroachment on the plant community and soil infiltration processes by contrasting vegetation and infiltration variables between three areas: a reference alpine meadow, an alpine meadow lightly encroached by smaller shrubs, and an alpine meadow heavily encroached by larger shrubs. We found that shrub encroachment reduced herbaceous plants aboveground biomass by 10–24 g m⁻² and root biomass by 1346–2239 g m⁻². The sedge functional group was gradually replaced by forbs. Initial infiltration rates gradually increased with shrub encroachment, and were approximately doubled and tripled at the earlier and later stages of encroachment, respectively. Overall, shrub encroachment improved the soil infiltrability by changing soil conditions and inhibiting herbaceous species due to the reduction of surface mastic epipedon and the formation of dead root channels. These findings suggest that the rapid transformation of alpine meadows into shrublands could have drastic effects on the surface runoff and stream-flow dynamics of the Qinghai-Tibetan Plateau.

1. Introduction

Alpine ecosystems are generally considered as “water towers” of watersheds due to their critical role in controlling runoff processes and aquifer recharge (McClymont et al., 2010). Alpine meadow is one of the typical vegetation community types on the Qinghai-Tibetan Plateau. The herbaceous plants in these communities form a thick turf and a root-dense mastic epipedon in the topsoil. This soil layer—which overlays an extensive permafrost—has high levels of biological activity and greatly impacts the hydrological processes and ecosystem functions (Wright

et al., 2009; Zhang et al., 2019). Alpine meadow's mastic epipedon is key for maintaining surface runoff while reducing sediment loss, which regulates surface water resource provision and soil erosion control (Niu et al., 2021). While in many regions surface runoff is seen as a negative process associated with land degradation (Wilcox and Huang, 2010), in other regions (including the Qinghai-Tibetan Plateau) this is a critical process that controls streamflow and groundwater recharge (Zou et al., 2014; Niu et al., 2021).

Shrub encroachment—the increase in abundance and dominance of native or exotic shrubs—has been documented worldwide over the last

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century. This phenomenon can be attributed to multiple factors including climate warming, alteration of natural fire regimes, higher atmospheric CO₂ concentrations, droughts, and higher nitrogen deposition (Kulmatiski and Beard, 2013). In alpine meadows, rising temperatures and the subsequent increase in nutrient availability are associated with shifts in the plant communities (Aguirre et al., 2021) and may be responsible for the increase in shrub abundance, as shown by long-term warming experiments (Walker et al., 2006; Sistla et al., 2013) and fertilization experiments (Shaver et al., 2001; Mack et al., 2004).

Shrub encroachment is a pervasive and often irreversible ecosystem transformation that can significantly alter biodiversity and ecosystem functions (Jackson et al., 2002; Knapp et al., 2008; Eldridge et al., 2011). At alpine meadows in particular, large-scale shrub encroachment has the potential to profoundly change the local vegetation community (Archer et al., 2017), thus threatening the effectiveness of these landscapes in maintaining high runoff and low sediment discharge rates (Niu et al., 2021). One of the main ways in which shrub encroachment might alter these hydrological processes is by changing the infiltration capacity of soils (Zou et al., 2014; Leite et al., 2020). This is a key property that controls runoff and soil moisture dynamics, and is involved in important vegetation-infiltration-soil moisture feedbacks (Schlesinger et al., 1990). Thus, it is essential to improve our understanding of the effects of shrub encroachment on the infiltration processes of alpine meadows—particularly those of Qinghai-Tibetan Plateau given their central role in the support of major environmental flows and in providing millions of people with clean water and other ecosystem services.

It is well known that changes in grassland community composition may affect root biomass and soil structure, and further affect soil infiltrability (Cui et al., 2019b). For example, plants with high aboveground biomass and abundant root systems may promote soil water and nutrient movement (Wang et al., 2016), and preferential water flow through root channels of shrubs or grasses enhances soil water infiltration (Tobella et al., 2014). Previous studies on shrub encroachment into alpine meadows have mostly focused on large regional scales (Li et al., 2016; Huang et al., 2018) and addressed changes in vegetation community and soil nutrients (Aguirre et al., 2021; Hao et al., 2021; Zhu et al., 2021). However, little is known about the quantitative relationship between shrub encroachment and the changes in infiltration processes in these ecosystems, and how these relate to changes in the herbaceous composition.

Encroaching shrubs are often present in different sizes, with the overall shrub coverage (i.e. stage of shrub encroachment) being directly reflected in the increasing shrubs' sizes and canopy areas. In the alpine meadows of the Qinghai-Tibetan Plateau for example, the influence of shrubs on understory vegetation growth was found to shift from facilitation to competition with increasing shrub canopy areas (Liu et al., 2020a). Therefore, to quantify the impacts of shrub encroachment on the infiltration capacity of alpine meadow soils, both shrub coverage and individual canopy areas must be used to assess the stage of shrub encroachment. For our study, we selected three areas that represent a chronosequence of shrub encroachment stages in the hinterland of the Qinghai-Tibetan Plateau, including (i) a typical alpine meadow with no shrub growth, (ii) an alpine meadow lightly encroached by mainly small shrubs, and (iii) an alpine meadow heavily encroached by comparatively larger shrubs (based on the shrub coverage and single canopy area) to represent the stages of shrub encroachment in the hinterland of Qinghai-Tibetan Plateau. The main goal of our research was to examine the aboveground and belowground changes in the plant community and the corresponding changes in soil infiltration capacity of alpine meadows undergoing shrub encroachment. We hypothesized that shrubs would inhibit herbaceous growth and change soil conditions in ways that enhance soil infiltration capacity, particularly at the heavily encroached alpine meadow.

2. Materials and methods

2.1. Study site

This study was conducted at alpine meadows in the Dari county of the Qinghai Province, China (32°36'42"-34°15'20"N, 98°15'29"-100°32'41"E). The site is located in the south-east region of the Tibetan Plateau at elevations above 4000 m a.s.l. (Fig. 1). The average annual air temperature is -1.2 °C, ranging from -12.9 °C in January to 9.1 °C in July. The average annual precipitation is ranges between 486.9 and 666.5 mm and is concentrated within the growing season (from May to September). The mean annual potential evapotranspiration is 1119.1 mm, and the annual total radiation flux ranges between 623.8 and 629.9 kJ·cm⁻² (Wang et al., 2005).

The main soil of the study site is classified as alpine meadow soil, and the native vegetation is mainly dominated by *Kobersia pygmaea* and *Elymus nutans*. The site has not experienced any form of land degradation (e.g. overgrazing). The surface layer of alpine meadow soil (0–10 cm) is well developed with a mixture of root masses and clay, and has very low bulk density. Two common shrubs (*Salix cupularis* and *Potentilla fruticosa* L.) have been encroaching into these areas over the recent decades (Han and Brierley, 2020). Plant growth is intimately related to climatic conditions (Wang et al., 2020), and because herbaceous plants at our site face a cold and harsh climate, they are generally dwarfed. Shrubs are also small and sparsely distributed during the initial stages of shrub encroachment, but as the process advances their canopies grow taller and wider, covering increasingly larger areas (Fig. 1).

2.2. Vegetation survey

A representative alpine meadow hillslope showing a typical shrub encroachment front was selected in 2020. Based on nine field surveys using large plots (5 m × 5 m), we selected three areas with different shrub coverage representing a catena of shrub encroachment gradients. These included: (i) a reference alpine meadow (AM; no shrubs growth); (ii) a lightly encroached alpine meadow with mainly small shrubs (AM_{LS}; average shrub coverage: 17.3%), and (iii) a heavily encroached alpine meadow dominated by larger shrubs (AM_{HS}; average shrub coverage: 56.7%). In each plot, a 50 cm × 50 cm small quadrat was randomly selected to quantify the changes in vegetation characteristics and soil parameters (soil aggregate structure, soil water holding capacity, and soil infiltration capacity). We measured the canopy area (assuming an ellipse) of individual shrubs in each plot, which revealed that at AM_{LS} and AM_{HS} the average canopy area was 0.15 m² and 0.87 m², respectively, while shrub height was 44 cm and 91 cm, respectively. The aboveground herbaceous parts of each species were harvested, oven dried at 75 °C for 48 h, weighed, and recorded as herbaceous biomass. Herbaceous coverage and the total count of each herbaceous species were also recorded.

2.3. Infiltration measurement

No rainfall occurred for at least three days prior to infiltration measurements. After carefully clipping vegetation at ground level, three soil infiltration measurements were conducted in each plot using a single-ring infiltrometer. A steel ring with 15-cm diameter and 15-cm height was gently inserted about 10 cm deep into the soil using a rubber hammer. At first, water was added into the ring as quickly as possible until reaching the ring edges (i.e. a 50 mm head). The time required for the whole volume to infiltrate was recorded, and the process was repeated until three consecutive infiltration times remained unchanged (indicating that infiltration rates reached a steady state). Then, infiltration rates were calculated as following (Zhang et al., 2017):

$$i = h / \Delta t$$

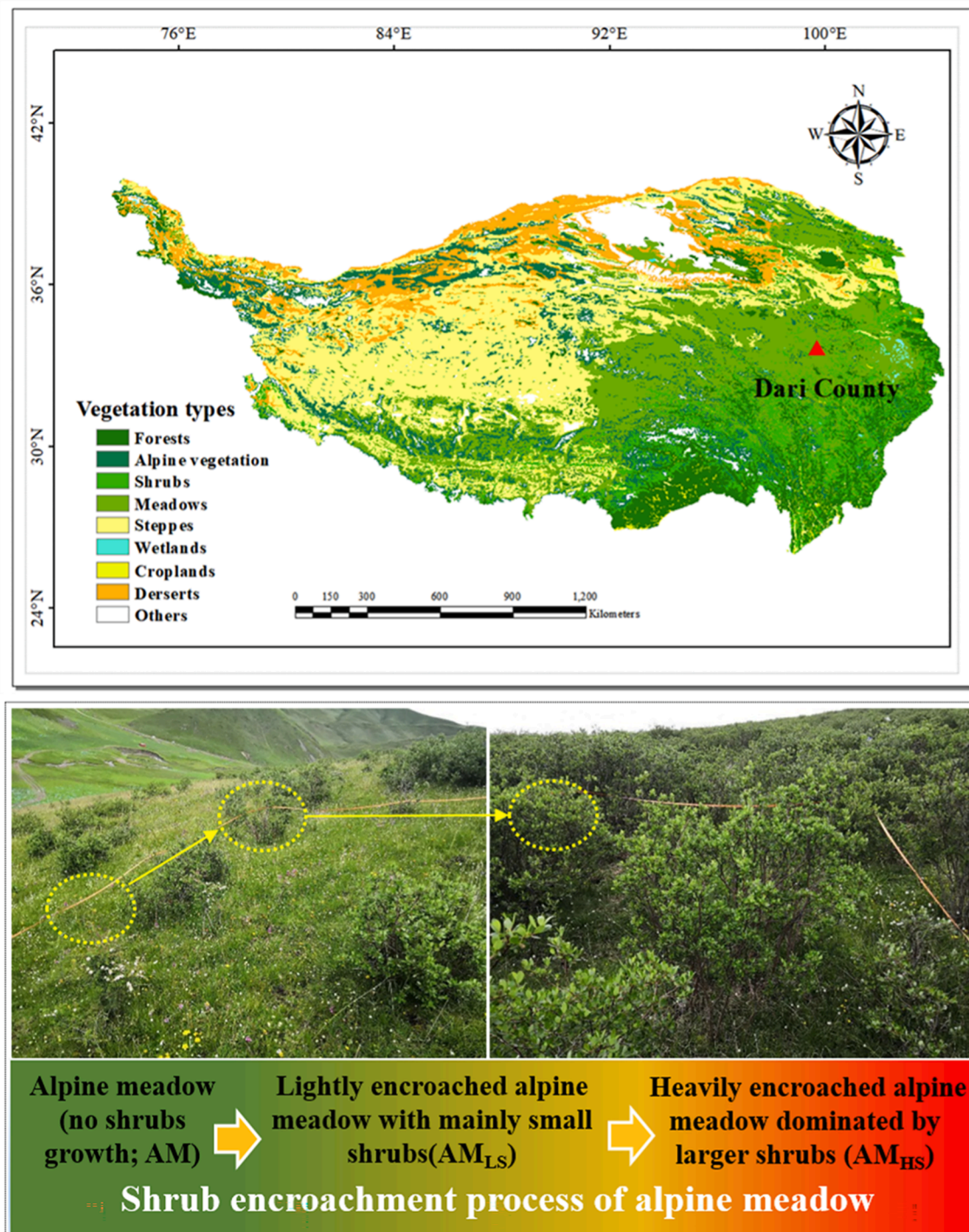


Fig. 1. Location of the study area in the Dari county, Qinghai Province, China. The photos below the map represent the shrub encroachment process of alpine meadow, including the alpine meadow, the lightly encroached alpine meadow with mainly small shrubs (AM_{LS}) and the heavily encroached alpine meadow dominated by larger shrubs (AM_{HS}).

where i represents the infiltration rate (mm h^{-1}), h represents the water head drop (mm), and Δt represents the time required for the water level to drop (hour).

According to Liu et al. (2020b), the head drop during the first three minutes of the infiltration tests was considered to represent the initial infiltration rate (IIR), while last three infiltrated heads were used to calculate the steady infiltration rate (SIR).

Maximum infiltration depth was determined following the staining method described in Leite et al. (2020). For that, a fixed amount of Brilliant blue dye was mixed in the water used in the infiltration tests

and soil profiles were excavated to measure infiltration depth. However, because of an impervious mixture of root masses and clays at approximately 10 cm deep, water did not penetrate past that layer and the average infiltration depth was about 10 cm at all plots (Fig. 3a).

2.4. Soil and root biomass sampling

After infiltration measurements, we collected three loose soil samples (approximately 1000 cm^3) at 10 cm increments from the surface to the 30 depth. These were washed to separate soil from herbaceous roots,

which were oven dried at 75 °C for 48 h to determine root biomass. At each plot, we also collected three fixed-volume samples at those same depths (0–10 cm, 10–20 cm and 20–30 cm) using a cylindrical steel ring. These were used to determine the soil total porosity (TP, %), soil capillary porosity (CP, %), and soil non-capillary porosity (NCP, %). The steel cylindrical rings with the samples were placed in a flat container after removing the upper covers. Water was then poured into the flat container until reaching the top edge of the steel rings. After absorbing water for 48 h, the bottom covers were removed and the steel cylindrical rings were placed for 2 h in a flat container filled with dry sand and weighed as M_1 (g). Finally, the samples (with rings included) were oven dried at 105 °C for 72 h and weighed as M_2 (g). The equations were the following (Cui et al., 2019a):

$$BD = \frac{M_2 - M}{V}$$

$$TP = \left(1 - \frac{BD}{ds}\right) \times 100$$

$$CP = \frac{(M_1 - M) - (M_2 - M)}{(M_2 - M)} \times \frac{BD}{V} \times 100$$

$$NCP = TP - CP$$

where BD is the bulk density of root-soil mixture (g cm^{-3}), ds is the soil particle density (we used the commonly accepted value of 2.65 g cm^{-3}), V is the sample volume (cm^3), and M is the weight of the steel cylindrical ring (g).

In each quadrat, additional undisturbed soil samples were collected at the 0–10 cm, 10–20 cm and 20–30 cm soil depth intervals to measure the distribution of soil aggregates following routine dry and wet sieving methods (Cui et al., 2019a). The mean weight diameter (MWD) and geometric mean diameter (GMD) of aggregates were calculated using the following equations:

$$MWD = \sum_{i=1}^n X_i W_i$$

$$GMD = \exp \left[\left(\sum_{i=1}^n W_i \lg X_i \right) / \left(\sum_{i=1}^n W_i \right) \right]$$

where X_i is the average diameter of the openings of two consecutive sieves; and W_i is the weight ratio of the aggregates remained on the i^{th} sieve.

Soil compactness (kg cm^{-2}) and volumetric soil moisture (%) at the 0–10 cm, 10–20 cm, and 20–30 cm soil depth intervals were measured adjacently to the locations where infiltration tests were performed using a soil compactness sensor (model) and a soil moisture probe (RR-7760H), respectively.

2.5. Statistical analyses

We used one-way analysis of variance (ANOVA) followed by least significant difference (LSD) *post hoc* test to assess significant differences in vegetation parameters (herbaceous aboveground biomass, plant functional composition, and herbaceous root biomass at the 0–10 cm, 10–20 cm and 20–30 cm soil depth intervals) between study areas (AM, AM_{LS} and AM_{HS}), as well as significant differences in soil parameters (soil compactness, soil moisture, GMD, MWD, TP, CP, and NCP) at the 0–10 cm, 10–20 cm and 20–30 cm soil depth intervals. These tests were performed using the SPSS 19.0 software (SPSS Inc., Chicago, IL, USA), with significance observed at the 0.05 confidence level.

Due to the close relationship between herbaceous aboveground biomass, herbaceous root biomass and soil parameters, two redundancy analyses (RDA) were conducted in this study. One was to test the correlations between soil infiltration rates (IIR and SIR) and vegetation

parameters (herbaceous aboveground biomass and herbaceous root biomass at the three soil depth intervals), and the other was to test the correlations between soil infiltration rates (IIR and SIR) and soil parameters (soil compactness, soil moisture, GMD, MWD, TP, CP, and NCP) at the three soil depth intervals. These analyses were conducted with the community ecology package *vegan* for C or R (Oksanen et al., 2007).

Moreover, we conducted principal components analysis (PCA) to explore the relations between soil parameters (soil compactness, soil moisture, GMD, MWD, TP, CP, and NCP) and reduce them down to a smaller number of principal components. The data was confirmed to be suitable for PCA according to the results from the Kaiser-Meyer-Olkin test (0.853) and the Bartlett's test of sphericity ($\chi^2_{\text{sqr}} = 289.5$, $df = 21$, p less than 0.0001). Based on the method described by T aylora and Tibshirani (2015), the most critical parameters determined from principal component analysis (PC-1 and PC-2) were used for the determination of soil parameters score as follows (Table 2; Fig. 5):

$$Y_{\text{SPP}} = (81.51Y_{\text{PC-1}} + 12.53Y_{\text{PC-2}})/93.04$$

where Y_{SPP} is the score of soil parameters. $Y_{\text{PC-1}}$ and $Y_{\text{PC-2}}$ are the score of PC-1 and PC-2, respectively. These tests were conducted in Origin 2021b software.

3. Results

3.1. Shrub encroachment changed community composition and reduced herbaceous biomass in alpine meadow

During the shrub encroachment process, the aboveground biomass of herbaceous species, which included three functional groups (sedges, legumes, and forbs), decreased significantly (by 10.84 g m^{-2} in AM_{LS} and 24.03 g m^{-2} in AM_{HS}) compared to AM (Table 1; Fig. 2a). Among functional groups, sedges biomass decreased by 15% and 19% in AM_{LS} and AM_{HS}, respectively, while the biomass of forbs increased by 13% and 16% in AM_{LS} and AM_{HS}, respectively. We also found that herbaceous roots were concentrated at the 0–10 soil depth interval (which contained 75–87% of the root biomass), and more important, that this biomass greatly decreased (by 1253.38 g m^{-2} at AM_{LS} and 2169.07 g m^{-2} at AM_{HS}) during the shrub encroachment process (Fig. 2b).

3.2. Changes in infiltration following shrub encroachment into alpine meadow

As it is expected, soil infiltration rates were highest at initial stages and then gradually decreased until reaching the stable state (Fig. 3a). Initial infiltration rates significantly increased with increasing levels of encroachment, and were 837 mm h^{-1} , 527.6 mm h^{-1} , and 253.0 mm h^{-1} at AM_{HS}, AM_{LS}, and AM, respectively (Fig. 3b). Conversely, steady infiltration rates were only significantly higher for the area at advanced stage of shrub encroachment (AM_{HS}, 396.5 mm h^{-1}), while there was no significant difference between the other two areas (80.2 mm h^{-1} and 116.2 mm h^{-1} at AM_{LS} and AM, respectively).

3.3. Infiltration was promoted by changes in topsoil parameters and root biomass

The RDA results showed that soil compactness, MWD, and herbaceous root biomass of the 0–10 cm soil layer were the parameters explained most of the variability in the initial infiltration rates across all areas. Conversely, steady infiltration rates were mainly explained by soil moisture and herbaceous root biomass at the 10–20 cm and 20–30 cm soil depths (Fig. 4).

The soil parameters score showed that shrub encroachment enhanced some soil conditions at the early stage of shrub encroachment processes (the score increased by 445.45% in AM_{LS} in relation to AM).

Table 1

Mean $\pm$ standard error values of the herbaceous aboveground biomass, herbaceous root biomass, soil moisture, and soil physical parameters (soil compactness; GMD, geometric mean diameter; MWD, mean weight diameter; TP, total soil porosity; CP, capillary porosity; NCP, non-capillary porosity) at the soil depth of 0–30 cm in the alpine meadow, the lightly encroached alpine meadow with mainly small shrubs (AM_{LS}) and the heavily encroached alpine meadow dominated by larger shrubs (AM_{HS}). For each soil depth but different areas, the values followed by a different letter are significantly different at 0.05 level.

Shrub encroachment process of alpine meadow	Soil depth (cm)	Herbaceous aboveground biomass (g m ⁻²)	Herbaceous root biomass (g m ⁻²)	Soil compactness (kg cm ⁻²)	Soil moisture (%)	GMD (mm)	MWD (mm)	TP (%)	CP (%)	NCP (%)
AM	0–10	156.51 $\pm$ 10.14b	3541.27 $\pm$ 20.35	20.38 $\pm$ 1.26b	51.03 $\pm$ 0.39c	1.54 $\pm$ 0.02a	3.47 $\pm$ 0.03a	81.53 $\pm$ 2.95	50.55 $\pm$ 3.83	30.97 $\pm$ 6.73
	10–20		327.57 $\pm$ 12.77	27.46 $\pm$ 1.75b	51.29 $\pm$ 0.20b	1.31 $\pm$ 0.02a	2.62 $\pm$ 0.02a	70.45 $\pm$ 1.10	57.79 $\pm$ 2.33	12.66 $\pm$ 2.42
	20–30		210.49 $\pm$ 8.66	32.63 $\pm$ 2.11b	47.71 $\pm$ 2.00b	1.26 $\pm$ 0.03b	2.61 $\pm$ 0.13b	67.56 $\pm$ 4.23	52.34 $\pm$ 4.44	15.22 $\pm$ 7.12
AM _{LS}	0–10	146.12 $\pm$ 16.09ab	2305.92 $\pm$ 5.51	13.50 $\pm$ 1.06a	48.45 $\pm$ 0.96b	1.67 $\pm$ 0.02b	3.93 $\pm$ 0.04c	79.14 $\pm$ 3.41	50.68 $\pm$ 3.51	28.46 $\pm$ 6.90
	10–20		249.43 $\pm$ 3.03	18.93 $\pm$ 1.66a	52.23 $\pm$ 0.29b	1.68 $\pm$ 0.01b	4.06 $\pm$ 0.05b	68.80 $\pm$ 0.63	56.82 $\pm$ 1.88	11.99 $\pm$ 1.37
	20–30		177.55 $\pm$ 8.23	24.22 $\pm$ 2.07a	49.99 $\pm$ 0.66b	1.12 $\pm$ 0.02a	2.12 $\pm$ 0.14ab	64.20 $\pm$ 2.53	52.69 $\pm$ 2.32	11.51 $\pm$ 1.62
AM _{HS}	0–10	132.59 $\pm$ 44.38a	1375.09 $\pm$ 45.46	15.83 $\pm$ 2.18ab	42.63 $\pm$ 1.02a	1.64 $\pm$ 0.03b	3.68 $\pm$ 0.05b	78.19 $\pm$ 0.30	52.83 $\pm$ 1.78	25.36 $\pm$ 2.05
	10–20		332.12 $\pm$ 27.55	18.52 $\pm$ 2.73a	43.95 $\pm$ 1.40a	1.32 $\pm$ 0.02a	2.74 $\pm$ 0.06a	71.32 $\pm$ 0.93	57.24 $\pm$ 1.75	14.08 $\pm$ 2.45
	20–30		132.72 $\pm$ 18.59	25.97 $\pm$ 2.83ab	42.7 $\pm$ 1.42a	1.11 $\pm$ 0.04a	1.95 $\pm$ 0.17a	65.76 $\pm$ 3.86	54.65 $\pm$ 3.08	11.10 $\pm$ 2.98

Table 2

Principal component analysis of the investigated soil parameters of alpine meadow during the shrub encroachment process. The parameters involved include soil compactness, soil moisture, geometric mean diameter of aggregates, mean weight diameter of aggregates, total soil porosity, capillary porosity, and non-capillary porosity.

Component	Initial eigenvalue			Extraction sums of squared loadings		
	Total	Percentage of Variance (%)	Cumulative (%)	Total	Percentage of Variance (%)	Cumulative (%)
1	5.7054	81.50573	81.50573	5.7054	81.50573	81.50573
2	0.87745	12.53502	94.04075	0.87745	12.53502	94.04075
3	0.16916	2.41663	96.45738			
4	0.1287	1.83861	98.29599			
5	0.07842	1.12036	99.41634			
6	0.02769	0.39561	99.81196			
7	0.01316	0.18804	100			

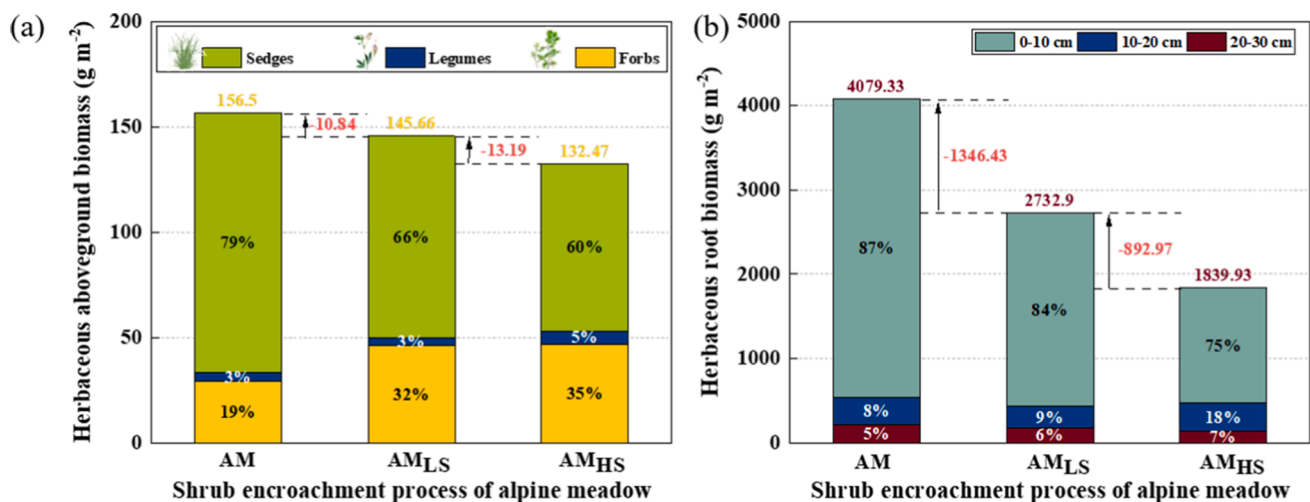


Fig. 2. The herbaceous aboveground biomass (a) and herbaceous root biomass (b) of the alpine meadow, the lightly encroached alpine meadow with mainly small shrubs (AM_{LS}) and the heavily encroached alpine meadow dominated by larger shrubs (AM_{HS}). The numbers at the top of columns represent the herbaceous aboveground biomass and herbaceous root biomass at 0–30 cm soil depth. The numbers inside the columns represent the proportion of herbaceous aboveground biomass in different plant functional groups (sedges, legumes and forbs). The numbers inside the columns represent the proportion of herbaceous root biomass in different soil depths (0–10 cm, 10–20 cm and 20–30 cm).

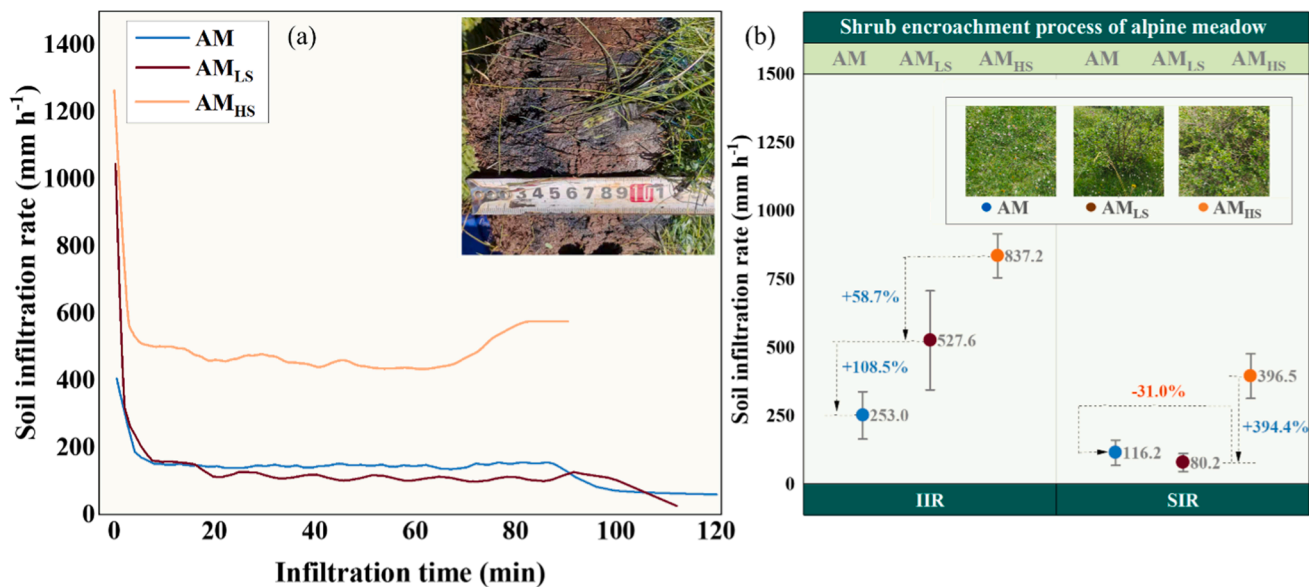


Fig. 3. Soil water infiltration process (a) and soil infiltration rates (mean $\pm$ se) at different stages (b) in the alpine meadow, the lightly encroached alpine meadow with mainly small shrubs (AM_{LS}) and the heavily encroached alpine meadow dominated by larger shrubs (AM_{HS}). The IIR and SIR refers to the initial infiltration rate and steady infiltration rate, respectively. The gray numbers near the dots represent the mean values. The average infiltration depth was about 10 cm determined by the staining method.

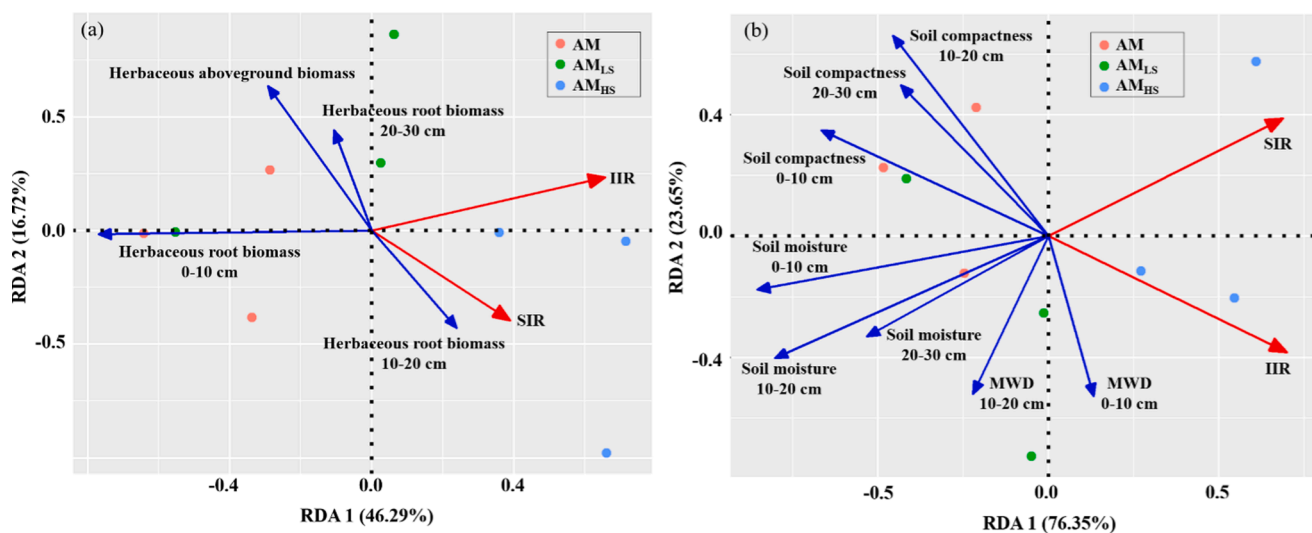


Fig. 4. Statistical results of the redundancy analysis (RDA) between infiltration rates and herbaceous biomass (a) and soil parameters (b). The red arrows indicate response variables and the blue arrows indicate explanatory variables. MWD, IIR and SIR refers to the mean weight diameter of aggregates, initial infiltration rate and steady infiltration rate, respectively. The AM, AM_{LS} and AM_{HS} refers to the alpine meadow, the lightly encroached alpine meadow with mainly small shrubs and the heavily encroached alpine meadow dominated by larger shrubs, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

However, the score was lower at the advanced stage of encroachment (decreased by 145.45% in AM_{HS} in relation to AM), which can be mainly attributed to the much lower soil moisture at that area (Fig. 5).

4. Discussion

Our results confirmed that shrub encroachment—a major landscape transformation happening in many of the world's grasslands and savannas—is dramatically altering the plant community and the soil functioning of alpine meadows in the Qinghai-Tibetan Plateau. Lower light availability under shrubs canopy has been shown to limit the growth of herbaceous plants in alpine meadows elsewhere (Martens et al., 2000; Aguirre et al., 2021). Additionally, as shrub cover increases they

intercept more rainfall and consume larger quantities of water (Asner et al., 2008; Normile, 2007), lowering soil moisture and limiting the survival of sedges. This was evidenced in our study by the replacement sedges with forb species as shrub encroachment becomes more severe. Following herbaceous species die-off (especially under the shaded canopies of shrubs), a large number of dead root channels are formed, thus promoting the high infiltration rates found for the encroached areas.

Such impacts of shrub encroachment on the herbaceous community are also reflected in the reductions of herbaceous root biomass and alteration of some soil properties. The reduction in root biomass is mainly seen at the top 10 cm of the soil, where most roots of the mattic epipedon are concentrated. The mattic epipedon is a special diagnostic surface horizon for soils in alpine meadow, and is mainly composed of

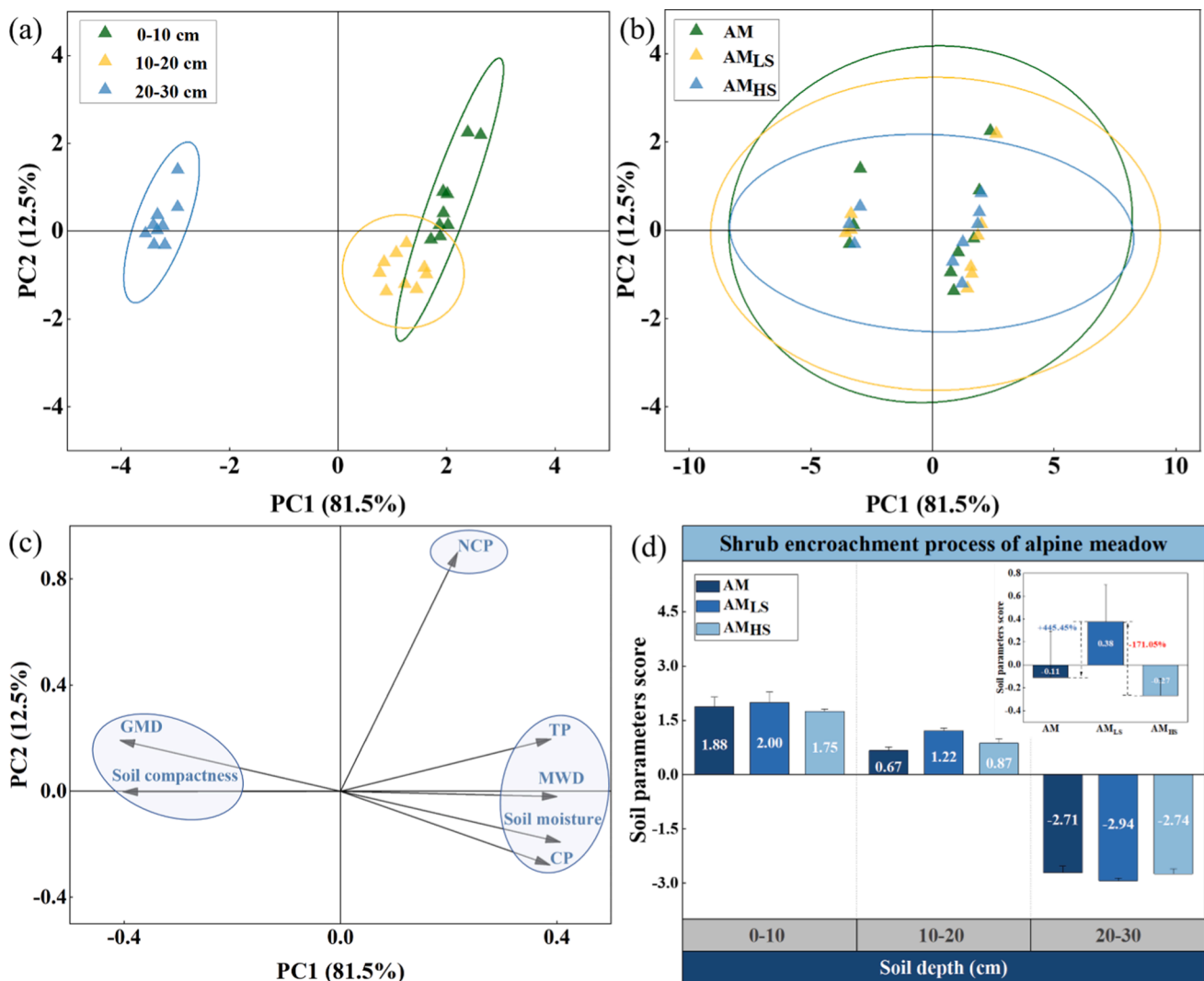


Fig. 5. The load of the principal component calculated by principal component analysis (a). Principal component scores across (b) different soil depths and (c) different treatments calculated by principal component analysis. Two PCA components explain 94.04% of the variance in soil parameters. GMD and MWD, TP, CP and NCP refer to the geometric mean diameter of aggregates, mean weight diameter of aggregates, total soil porosity, capillary porosity, and non-capillary porosity, respectively. Soil parameters score (mean $\pm$ se) at different soil depths of our three areas (d). The soil parameters score is calculated by principal component analysis. The numbers inside the columns represent the mean values. The AM, AM_{LS} and AM_{HS} refers to the alpine meadow, the lightly encroached alpine meadow with mainly small shrubs and the heavily encroached alpine meadow dominated by larger shrubs, respectively.

dense sedge roots. These account for about 65% of the roots in the whole soil profile, and are the main contribution to the high organic matter content, high water holding capacity, and low thermal conductivity that characterizes these soils (Biskop et al., 2016). Moreover, the decomposition of roots could increase the amount of organic matter and improve the formation and stability of soil aggregates, as evidenced by the higher GMD at the shrub encroached areas. Higher aggregate stability has been shown to promote infiltration through preferential flow pathways and increase the infiltrability of the relatively impermeable sub-surface of the matic epipedon (Dai et al., 2019). Additionally, direct reductions in soil moisture by shrubs leads to higher unsaturated hydraulic conductivity, which affects mainly the initial stages of the infiltration process. Such mechanisms—particularly changes in root biomass and soil moisture, which were highest at AM and lowest at AM_{HS}—could explain why IIR gradually increased with increasing levels of shrub encroachment.

Another important way in which shrubs promote infiltration is by directly enhancing macroporosity through both live and decaying woody roots (Beven and Germann, 2013), which is amplified with increasing shrub age and size (Leung et al., 2017). Therefore, the high steady infiltration rate at the lightly encroached alpine meadow with

mainly small shrubs, is probably explained by a combination of more stable aggregates and higher macroporosity. Since the heavily encroached alpine meadow dominated by larger shrubs has larger shrub cover, bigger shrubs, and has likely been experiencing encroachment for a longer time, these changes have accumulated more than at the lightly encroached alpine meadow with mainly small shrubs.

Overall, our results clearly support our initial hypothesis that shrubs would increase the soil infiltrability of alpine meadows, and are in agreement with multiple studies that have shown the same trend in other ecosystems (Eldridge et al., 2012; Leite et al., 2020). Within these other studies, there is a general consensus that the higher infiltrability brought by shrubs can lead to lower surface runoff rates (Wilcox and Huang, 2010). While this is often seen as a positive outcome associated with, for example, lower erosion and higher groundwater recharge, this is not the case for the Tibetan-Plateau. In this region, overland flow is the dominant source of streamflow, and surface runoff maintenance is considered a key ecosystem service provided by alpine meadows (Wu et al., 2020; Niu et al., 2021). Surface runoff occurs in two forms: saturation-excess overland flow and infiltration-excess overland flow. The first occurs when the soil profile is fully saturated (e.g. due to

perched water tables), while the later occurs when rainfall intensity exceeds the infiltration capacity of the soil. Our results suggest that shrub encroachment in the Qinghai-Tibetan Plateau could reduce the likelihood of both types of runoff: of saturation-excess overland flow by lowering soil moisture and of infiltration-excess overland flow by enhancing infiltration rates at both unsaturated (IIR) and saturated (SIR) stages. Such changes—combined with the higher interception rates and transpiration demand of encroaching shrubs—were found to significantly reduce streamflow in other regions, such as the Cross Timbers in the United States (Qiao et al., 2017; Zou et al., 2014).

Moreover, the dense matted epipedon of alpine meadows can form a protective cushion that alleviates livestock trampling (Lin et al., 2008). By reducing herbaceous cover and belowground biomass, shrub encroachment weakens this protective function and increases the risk of alpine meadow degradation. Such degradation is likely to promote soil erosion in the shrub interspaces and result in the concentration of water and nutrient resources in the shrub canopy zones (Schlesinger et al., 1990). The formation of such “islands of fertility” under encroaching shrubs has been documented in several other regions (Hibbard et al., 2001; Rodríguez-Iturbe et al., 2019) and can lead to reinforcing feedbacks that promote the persistence of the alternative, degraded state. Based on our findings and on the available literature, it might be concluded that by changing the surface water dynamics and negatively affecting the herbaceous vegetation, shrub encroachment (especially if combined with grazing) will likely initiate a degradation process in the alpine meadows of the Qinghai-Tibetan Plateau. Given the major importance of these ecosystems for the water security of millions of people, further investigations on this topic are urgently needed.

5. Conclusion

In agreement with our initial hypothesis, our results demonstrated that shrub encroachment dramatically increased soil infiltrability of alpine meadows in the Qinghai-Tibetan Plateau. At the early stage of shrub encroachment (when shrubs are smaller and cover less than 20% of the site), the herbaceous biomass is significantly altered and the initial infiltration rates are doubled. As the process advances (with mainly larger shrubs covering more than 50% of the site) these effects are accentuated, and both the initial and the steady state infiltration rates become 3–4 times higher than at the alpine meadow. In addition, we found that shrub encroachment altered the functional composition of the alpine meadow, with sedges getting replaced by forb species. The higher infiltration rates at shrub encroached areas can be attributed to multiple mechanisms including: (i) reductions in soil moisture due to higher evapotranspiration by shrubs, (ii) increased aggregate stability and porosity following the decomposition of the matted epipedon, and (iii) the development of preferential flow pathways along shrub roots. These changes in infiltration will potentially alter the surface runoff dynamics, especially as shrub encroachment becomes more severe. Given the importance of overland flow for sustaining streamflow in the Qinghai-Tibetan Plateau, a better understanding of the effects of shrub encroachment in the surface hydrology of alpine meadows must become a priority.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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